The documentation process conversion measures necessary to comply with this revision shall be completed by 3 September 1997

INCH POUND

MIL-PRF-19500/602B
3 June 1997
SUPERSEDING
MIL-S-19500/602A
21 January 1994

PERFORMANCE SPECIFICATION

SEMICONDUCTOR DEVICE, FIELD EFFECT RADIATION HARDENED (TOTAL DOSE ONLY)
TRANSISTORS, N-CHANNEL, SILICON TYPES 2N7265, 2N7266, AND 2N7267
JANTXVM, D, R, F, G, AND H AND JANSM, D, R, F, G, AND H

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for an N-Channel, enhancement-mode, MOSFET, Radiation Hardened (total dose only), power transistor intended for use in high density power switching applications. Two levels of product assurance are provided for each device type specified in MIL-PRF-19500, with Avalanche Energy Maximum rating (E_{AS}) and Maximum Avalanche Current (I_{AS}).

1.2 Physical dimensions. See figure 1, TO-204AE (2N7265, 2N7266) and TO-204AA (2N7267).

1.3 Maximum ratings. T_A = +25°C, unless otherwise specified.

Type	P_T 1/	P_T	V_{DS}	V_{DG}	V_{GS}	I_{D1} 2/	I_{D2}	I_S	I_{DM}	T_{op} and T_{STG}	V_{ISO} 70,000 ft. altitude
	T_C = +25°C	T_A = +25°C				T_C = +25°C	T_C = +100°C	2/			
	W	W	V dc	V dc	V dc	A dc	A dc	A dc	A (pk)	°C	V dc
2N7265	150	4	100	100	±20	34.0	21.0	34.0	136	-55	N/A
2N7266	150	4	200	200	±20	26.0	16.0	26.0	104	to	N/A
2N7267	150	4	500	500	±20	11.0	7.0	11.0	44	+150	500

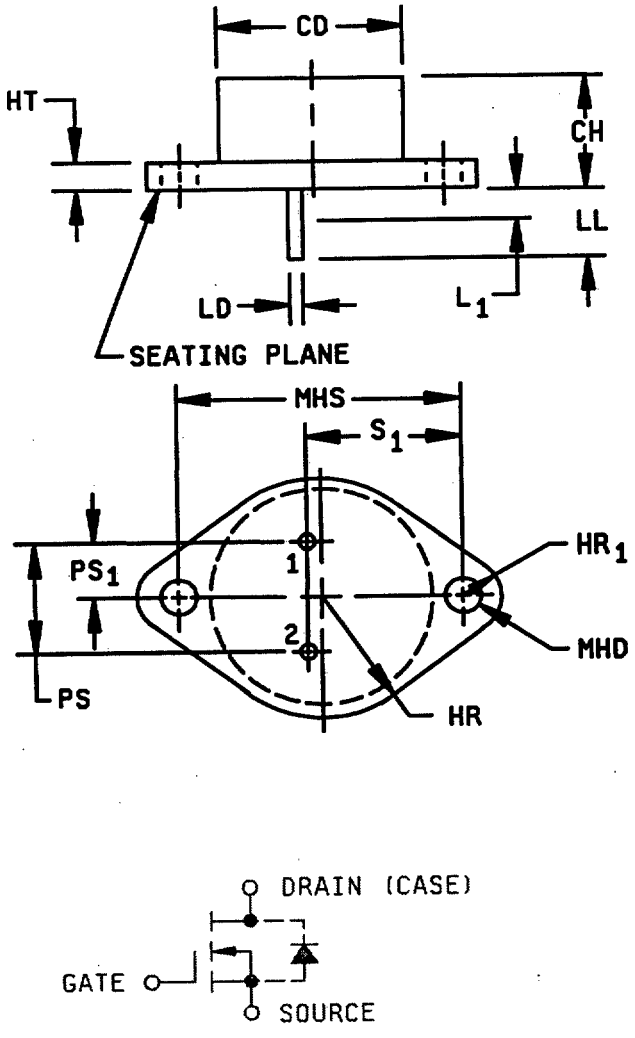
1/ Derate linearly 1.2 W/°C for T_C > +25°C;

2/ $$I_D = \sqrt{\frac{T_J \text{max} - T_C}{(R_{\theta JC}) x (r_{DSon} \text{ at } T_J \text{max})}}$$

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Supply Center Columbus, ATTN: DSCC-VAT, 3990 East Broad Street, Columbus, OH 43216-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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	Dimensions			
	Millimeters		Inches	
Symbol	Min	Max	Min	Max
CH	6.35	9.15	.250	.360
LD Note 3	1.45	1.60	.057	.063
LD Note 4	0.97	1.10	.038	.043
CD		22.23		.875
PS	10.67	11.18	.420	.440
PS_1	5.21	5.72	.205	.225
HT	1.52	3.43	.060	.135
LL	7.92	12.70	.312	.500
L_1		12.70		.500
MHD	3.84	4.09	.151	.161
MHS	29.90	30.40	1.177	1.197
HR	12.57	13.34	.495	.525
HR_1	3.33	4.78	.131	.188
S_1	16.64	17.15	.655	.675

NOTES:
1. Dimensions are in millimeters.
2. Inch equivalents are given for general information only.
3. These dimensions pertain to the TO-204AE (2N7265, 2N7266).
4. These dimensions pertain to the TO-204AA (2N7267).
5. Lead 1 Source; Lead 2 Gate; Drain is electrically connected to the case.

FIGURE 1. Physical dimensions for TO-204AE (2N7265, 2N7266) and TO-204AA (2N7267).

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1.4 Primary electrical characteristics at T_C = +25°C.

Type	Min $V_{(BR)DSS}$ V_{GS} = 0 I_D = 1.0 mA dc	$V_{GS(TH)1}$ $V_{DS} \geq V_{GS}$ I_D = 1.0 mA dc		Max I_{DSS1} V_{GS} = 0 V_{DS} = 80% of rated V_{DS}	Max $r_{DS(ON)}$ 1/ V_{GS} = 12 V dc		$R_{\theta JC}$ max	E_{AS} at I_{D1}	I_{AS}
					T_J = +25°C at I_{D2}	T_J = +150°C at I_{D2}			
	V dc	V dc		µA dc	ohm	ohm	°C/W	mJ	A
		Min	Max						
2N7265	100	2.0	4.0	25	0.065	0.132	0.83	500	34.0
2N7266	200	2.0	4.0	25	0.100	0.230	0.83	500	26.0
2N7267	500	2.0	4.0	50	0.450	1.260	0.83	500	11.0

1/ Pulsed (see 4.5.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated specifications or specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Qualification. Devices furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.3).

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500 and as follows.

I_{AS} - Rated avalanche current, nonrepetitive

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified on figure 1 herein.

3.3.1 Lead material and finish. Lead material shall be Kovar or Alloy 52; a copper core or plated core is permitted. Lead finish shall be solderable in accordance with MIL-PRF-19500 and herein. Where a choice of lead material or finish is desired, it shall be specified in the contract or purchase order (see 6.5).

3.3.2 Internal construction. Multiple chip construction is not permitted to meet the requirements of this specification.

3.4 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.5 Electrostatic discharge protection. The devices covered by this specification require electrostatic discharge protection.

3.5.1 Handling. MOS devices must be handled with certain precautions to avoid damage due to the accumulation of static charge. However, the following handling practices are recommended (see 3.5).

a. Devices should be handled on benches with conductive handling devices.

b. Ground test equipment, tools, and personnel handling devices.

c. Do not handle devices by the leads.

d. Store devices in conductive foam or carriers.

e. Avoid use of plastic, rubber, or silk in MOS areas.

f. Maintain relative humidity above 50 percent if practical.

g. Care should be exercised during test and troubleshooting to apply not more than maximum rated voltage to any lead.

h. Gate must be terminated to source, $R \leq 100$ k, whenever bias voltage is to be applied drain to source.

4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3)

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500.

4.2.1 Group E inspection. Group E inspection shall be conducted in accordance with MIL-PRF-19500, and table III herein.

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4.3 Screening (JANS, JANTX and JANTXV levels only). Screening shall be in accordance with MIL-PRF-19500 (Appendix E, table IV), and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see appendix E, table IV of MIL-PRF-19500)	Measurement	
	JANS level	JANTXV levels
1/	Method 3470 (see 4.5.4).	Method 3470 (see 4.5.4).
1/	Method 3161 (see 4.5.3).	Method 3161 (see 4.5.3).
1/	Gate stress test (see 4.5.5).	Gate stress test (see 4.5.5).
9 1/	Subgroup 2 of table I herein. I_{GSS}, I_{DSS1}	Subgroup 2 of table I herein.
10	MIL-STD-750, method 1042 test condition B.	MIL-STD750, method 1042 test condition B.
11	I_{GSSF1}, I_{GSSR1}, I_{DSS1}, $r_{DS(on)}$, $V_{GS(TH)}$ Subgroup 2 of table I herein. ΔI_{GSSF1} = ±20 nA dc or ± 100% of initial value, whichever is greater. ΔI_{GSSR1} = ± 20 nA dc or ± 100% of initial value, whichever is greater. ΔI_{DSS1} = ± 10 :A dc or ± 100% of initial value, whichever is greater.	I_{GSSF1}, I_{GSSR1}, I_{DSS1}, $r_{DS(on)}$, $V_{GS(th)}$ Subgroup 2 of table I herein.
12	MIL-STD-750, method 1042, test condition A.	MIL-STD-750, method 1042, condition A.
13	Subgroups 2 and 3 of table I herein. ΔI_{GSSF1} = ± 20 nA dc or ± 100% of initial value, whichever is greater. ΔI_{GSSR1} = ± 20 nA dc or ± 100% of initial value, whichever is greater. ΔI_{DSS1} = ± 10 µA dc or ± 100% of initial value, whichever is greater. $\Delta r_{DS(on)1}$ = ± 20% of initial value. $\Delta V_{GS(th)1}$ = ± 20% of initial value.	Subgroups 2 and 3 of table I herein. ΔI_{GSSF1} = ± 20nA dc or 100% of initial value, whichever is greater. ΔI_{GSSR1} = ± 20nA dc or 100% of initial value, whichever is greater. ΔI_{DSS1} = ± 0 µA dc or 100% of initial value, whichever is greater. $\Delta r_{DS(on)1}$ = ± 20% of initial value. $\Delta V_{GS(th)1}$ = ± 20% of initial value.

1/ Shall be performed anytime before screen 10.

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4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500, and as specified herein. Alternate flow is allowed for quality conformance inspection in accordance with appendix E of MIL-PRF-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with appendix E, table V of MIL-PRF-19500 and table I herein. End-point electrical measurements shall be in accordance with table I, group A, subgroup 2 herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table VIa (JANS) and table VIb (JANTX and JANTXV) of MIL-PRF-19500, and herein. Electrical measurements (end points) and delta requirements shall be in accordance with table I, group A, subgroup 2 herein.

4.4.2.1 Group B inspection, appendix E, table VIa (JANS) of MIL-PRF-19500.

Subgroup	Method	Condition
B3	2075	See 3.3.2
B3	2077	SEM qualification may be performed anytime prior to lot formation.
B4	1042	Condition D, 2,000 cycles. No heat sink nor forced-air cooling on the device shall be permitted during the on cycle. The heating cycle shall be 30 seconds minimum.
B5	1042	Test condition A, V_{DS} = rated T_A = +175°C, t = 120 hours. Read and record $V_{(BR)DSS}$ (pre and post) at I_D = 1 mA. Read and record I_{DSS} (pre and post) in accordance with table I, group A, subgroup 2. No more than 15% of the sample shall be permitted to have a $\Delta V_{BR(DSS)}$ shift of more than 10% and ΔI_{DSS} greater than 25 µA.
B5	1042	Condition B, V_{GS} = rated; T_A = +175°C; t = 24 hours.
B5	2037	Bond strength (Al-Au die interconnects only); test condition A.
B6	3161	See 4.5.2.

4.4.2.2 Group B inspection, appendix E, table VIb (JANTX and JANTXV) of MIL-PRF-19500.

Subgroup	Method	Condition
B2	1051	Test condition G, 25 cycles.
B3	1042	Test condition D, 2,000 cycles; The heating cycle shall be 30 seconds minimum.
B4	2075	See 3.3.2
B4	2037	Test condition A. All internal bond wires for each device shall be pulled separately.
B5 & B6	----	Not applicable

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4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table VII of MIL-PRF-19500 and as follows. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable of table I, group A, subgroup 2 herein.

Subgroup	Method	Condition
C2	1056	Test condition B.
C2	2036	Test condition A; weight = 10 pounds; t = 15 s.
C2	1021	Omit initial conditioning.
C5	1001	Test condition C. For device type 2N7267: V_{DS} = 500 V; $I_{(ISO)}$ < 0.25 mA.
C6	1042	Test condition D, 6,000 cycles. The heating cycle shall be 30 seconds minimum.

4.4.4 Group D Inspection. Group D inspection shall be conducted in accordance with appendix E, table VIII of MIL-PRF-19500 and table II herein. Level F and G represent 300 k and 600 k rad (Si) respectively.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in of MIL-STD-750.

4.5.2 Thermal resistance. Thermal resistance measurements shall be performed in accordance with method 3161 of MIL-STD-750. The maximum limit of $R_{\theta JC(max)}$ = 0.83°C/W. The following parameter measurements shall apply:

a.	Measuring current (I_M) -------	10 mA.
b.	Drain heating current (I_H) ------	4 A.
c.	Heating time (t_H) ----------	Steady-state (see MIL-STD-750, method 3161 for definition).
d.	Drain-source heating voltage (V_H) --	25 V.
e.	Measurement time delay (t_{MD}) ----	30 to 60 µs.
f.	Sample window time (t_{SW}) ------	10 µs maximum.

4.5.3 Thermal impedance ($Z_{\theta JC}$ measurements). The $Z_{\theta JC}$ measurements shall be performed in accordance with MIL-STD-750, method 3161. The maximum limit (not to exceed figure 3, thermal impedance curves and the group A, subgroup 2 limits) for $Z_{\theta JC}$ in screening (appendix E, table IV of MIL-PRF-19500) shall be derived by each vendor by means of statistical process control. When the process has exhibited control and capability, the capability data shall be used to establish the fixed screening limit. In addition to screening, once a fixed limit has been established, monitor all future sealing lots using a random five piece sample from each lot to be plotted on the applicable X, R chart. If a lot exhibits an out of control condition, the entire lot shall be removed from the line and held for Engineering evaluation and disposition. This procedure shall be used in lieu of an in line monitor.

a.	Measuring current (I_M) ----------	10 mA.
b.	Drain heating current (I_H) ----------	4 A minimum.
c.	Heating time (t_H) ----------	100 ms.
d.	Drain-source heating voltage (V_H) ----------	25 V.
e.	Measurement time delay (t_{MD}) ----------	30 to 60 µs.
f.	Sample window time (t_{SW}) ----------	10 µs maximum.

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4.5.4 Single pulse avalanche energy E_{AS}.

a.	Peak current (I_{AS})	$I_{AS(max)}$
b.	Peak gate voltage (V_{GS})	10 V.
c.	Gate to source resistor (R_{GS})	$25\Omega \leq R_{GS} \leq 200\Omega$.
d.	Initial case temperature (T_C)	+25°C +10°C, -5°C.
e.	Inductance (L)	$[2E_{AS}/(I_{D1})^2] [(V_{BR} - V_{DD})]$ mH minimum $\frac{}{V_{BR}}$
f.	Number of pulses to be applied	1 pulse minimum.
g.	Supply voltage (V_{DD})	25 V for 2N7265 50 V for 2N7266 and 2N7267

4.5.5 Gate stress test.

a. V_{GS} = 24 V minimum.

b. t = 250 µs minimum.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical inspection	2071					
Subgroup 2						
Thermal impedance 2/	3161	See 4.5.3	$Z_{\theta JC}$		0.65	°C/W
Breakdown voltage, drain to source	3407	V_{GS} = 0 V; I_D = 1 mA dc Bias condition C	$V_{(BR)DSS}$			
2N7265				100		V dc
2N7266				200		V dc
2N7267				500		V dc
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$ I_D = 1 mA dc	$V_{GS(TH)1}$	2.0	4.0	V dc
Gate current	3411	V_{GS} = +20 and -20 V dc Bias condition C, V_{DS} = 0	I_{GSS1}		±100	nA dc
Drain current	3413	V_{GS} = 0 V dc, Bias condition C V_{DS} = 80% of rated V_{DS}	I_{DSS1}			
2N7265					25	µA dc
2N7266					25	µA dc
2N7267					50	µA dc
Static drain to source "ON"-state resistance	3421	V_{GS} = 12 V dc, condition A Pulsed (see 4.5.1), I_D = I_{D2}	$r_{DS(on)1}$			
2N7265					0.065	ohm
2N7266					0.100	ohm
2N7267					0.450	ohm
Static drain to source "ON"-state resistance	3421	V_{GS} = 12 V dc, condition A Pulsed (see 4.5.1), I_D = I_{D1}	$r_{DS(on)2}$			
2N7265					0.070	ohm
2N7266					0.110	ohm
2N7267					0.500	ohm
Forward voltage	4011	Pulsed (see 4.5.1), I_D = I_{D1} V_{GS} = 0 V dc	V_{SD}			
2N7265					1.4	V
2N7266					1.4	V
2N7267					1.6	V

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 3						
High temperature operation		$T_C = T_J = +125°C$				
Gate current	3411	V_{GS} = +20 and -20 V dc Bias condition C, V_{DS} = 0	I_{GSS2}		± 200	nA dc
Drain current	3413	V_{GS} = 0 V dc, Bias condition C V_{DS} = 100% of rated V_{DS}	I_{DSS2}		1.0	mA dc
		V_{DS} = 80% of rated V_{DS}	I_{DSS3}		0.25	mA dc
Static drain to source "ON"-state resistance	3421	V_{GS} = 12 V dc Pulsed (see 4.5.1), $I_D = I_{D2}$	$r_{DS(on)3}$			
2N7265					0.132	ohm
2N7266					0.200	ohm
2N7267					1.000	ohm
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$, I_D = 1 mA dc	$V_{GS(TH)2}$	1.0		V dc
Low temperature operation		$T_C = T_J = -55°C$				
Gate to source voltage(threshold)	3403	$V_{DS} \geq V_{GS}$, I_D = 1 mA dc	$V_{GS(TH)3}$		5.0	V dc
Subgroup 4						
Forward Transconductance	3475	I_D = rated I_{D2}, V_{DD} = 15 V (See 4.5.1)	g_{FS}			
2N7265				8.0		S
2N7266				8.0		S
2N7267				4.0		S
Switching time test	3472	$I_D = I_{D1}$, V_{GS} = 12 V dc R_G = 2.35Ω, V_{DD} = 50% of rated V_{DS}				

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 4 - Continued.						
Turn-on delay time			$t_{d(on)}$			
2N7265					45	ns
2N7266					33	ns
2N7267					45	ns
Rise time			t_r			
2N7265					190	ns
2N7266					140	ns
2N7267					190	ns
Turn-off delay time			$t_{d(off)}$			
2N7265					170	ns
2N7266					140	ns
2N7267					190	ns
Fall time			t_f			
2N7265					130	ns
2N7266					140	ns
2N7267					130	ns
Subgroup 5						
Safe operating area test (high voltage)	3474	See figures 3, 4, and 5 t_p = 10 ms minimum, V_{DS} = 80% of max rated V_{DS} ($V_{DS} \leq 200$)				
Electrical measurements		See table I, group A, subgroup 2				
Subgroup 6						
Not applicable						
Subgroup 7						
Gate charge	3471	Condition B				
On-state gate charge			$Q_{g(on)}$			
2N7265					160	nC
2N7266					170	nC
2N7267					150	nC

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 7 - Continued.						
Gate to source charge			Q_{gs}			
2N7265					35	nC
2N7266					30	nC
2N7267					30	nC
Gate to drain charge			Q_{gd}			
2N7265					65	nC
2N7266					60	nC
2N7267					75	nC
Reverse recovery time	3473	di/dt ≤ 100 A/µs $V_{DD} \leq 30$ V, $I_D = I_{D1}$	t_{rr}			
2N7265					570	ns
2N7266					820	ns
2N7267					1100	ns

1/ For sampling plan, see MIL-PRF-19500.

2/ This test is required for the following end-point measurements only (not intended for screen 13):

JANS - group B, subgroup 3 and 4

JANTX and JANTXV - group B, subgroup 2 and 3;
group C, subgroup 6;
group E, subgroup 1

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TABLE II. Group D inspection.

Inspection 1/ 2/ 3/	MIL-STD-750		Symbol	Preirradiation limits				Postirradiation limits				Unit
						3/				3/		
	Method	Conditions		M, D, and R		F, G, and H		M, D, and R		F, G, and H		
				Min	Max	Min	Max	Min	Max	Min	Max	
Subgroup 1												
Not applicable												
Subgroup 2		T_C = +25°C										
Steady-state total dose irradiation (V_{GS} bias) 4/	1019	V_{GS} = 12 V V_{DS} = 0										
Steady-state total dose irradiation (V_{DS} bias) 4/	1019	V_{GS} = 0 V_{DS} = 80% of rated VDS (pre-irradiation)										
End-point electricals												
Breakdown voltage, drain to source	3407	V_{GS} = 0; I_D = 1 mA Bias cond. C	V_{BRDSS}									
2N7265				100		100		100		100		V dc
2N7266				200		200		200		200		V dc
2N7267				500		500		500		500		V dc
Gate to source 5/ voltage (threshold)	3403	$V_{DS} \geq V_{GS}$ I_D = 1 mA	V_{GSth}									
2N7265				2	4	2	4	2	4	1.25	4.50	V dc
2N7266				2	4	2	4	2	4	1.25	4.50	V dc
2N7267				2	4	2	4	2	4	1.25	4.50	V dc
Gate current	3411	V_{GS} = 20 V V_{DS} = 0 Bias cond. C	I_{GSSF1}		100		100		100		100	nA dc
Gate current	3411	V_{GS} =-20 V V_{DS} = 0 Bias cond. C	I_{GSSR1}		-100		-100		-100		-100	nA dc

See footnotes at end of table.

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TABLE II. Group D inspection - Continued.

Inspection 1/ 2/ 3/	MIL-STD-750		Symbol	Preirradiation limits		Preirradiation limits 3/		Unit
				M, D, and R		F, G, and H		
	Method	Conditions		Min	Max	Min	Max	
Subgroup 2 - Continued.								
Drain current	3413	V_{GS} = 0 Bias cond. C V_{DS} = 80% of rated V_{DS} (pre-irradiation)	I_{DSS}					
2N7265					25		25	µA dc
2N7266					25		25	µA dc
2N7267					50		50	µA dc
Static drain to source on-state resistance	3421	V_{GS} = 12 V Cond. A pulsed see 4.5.1 $I_D = I_{D2}$	r_{DSon1}					
2N7265					0.065		0.065	Ω
2N7266					0.100		0.100	Ω
2N7267					0.450		0.450	Ω
Forward voltage source drain diode	4011	V_{GS} = 0 V $I_D = I_{D1}$	V_{SD}					
2N7265					1.4		1.4	V
2N7266					1.4		1.4	V
2N7267					1.6		1.6	V

1/ For sampling plan see MIL-PRF-19500.
2/ Group D qualification may be performed anytime prior to lot formation. Wafers qualified to these group D QCI requirements may be used for any other detail specification utilizing the same die design.
3/ The F designation represent devices which pass end-points at both 100 k and 300 k rads (Si). The G designation represents devices which pass 100 k, 300 k, and 600 k rad (Si) end-points.
4/ Separate samples shall be pulled for each bias.
5/ H must meet end-points for 300 k and 1000 k rad (Si).

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TABLE III. Group E inspection (all quality levels) for qualification only.

Inspection	MIL-STD-750		Qualification and large lot quality conformance inspection
	Method	Conditions	
Subgroup 1			12 devices c = 0
Thermal shock (temperature cycling)	1051	-55°C to +150°C, 500 cycles	
Hermetic seal	1071		
Fine leak			
Gross leak			
Electrical measurements		See table I, group A, subgroup 2	
Subgroup 2 1/			12 devices c = 0
Steady-state reverse bias	1042	Condition A: 1,000 hours	
Electrical measurements		See table I, group A, subgroup 2	
Steady-state gate bias	1042	Condition B: 1,000 hours	
Electrical measurements		See table I, group A, subgroup 2	
Subgroup 3			
Not applicable			
Subgroup 4			
Thermal resistance	3161	$R_{\theta JC}$ = 0.83 °C/W maximum. See 4.5.2	12 devices c = 0

1/ A separate sample for each test shall be pulled.

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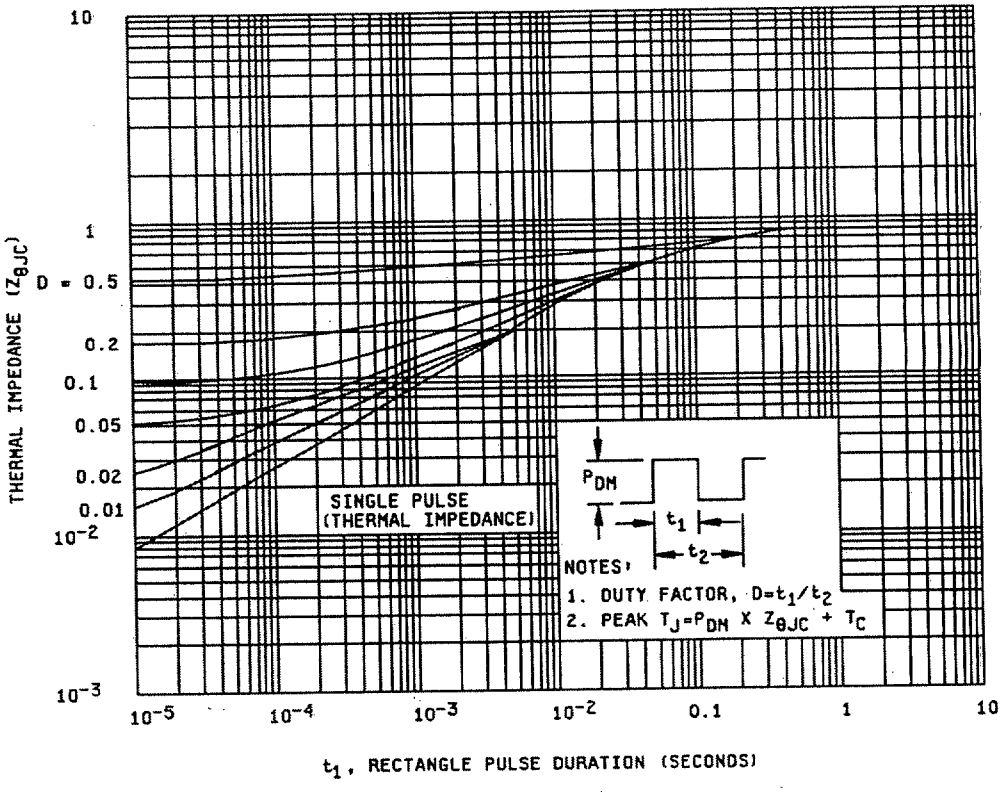


FIGURE 2. Thermal impedance curves.

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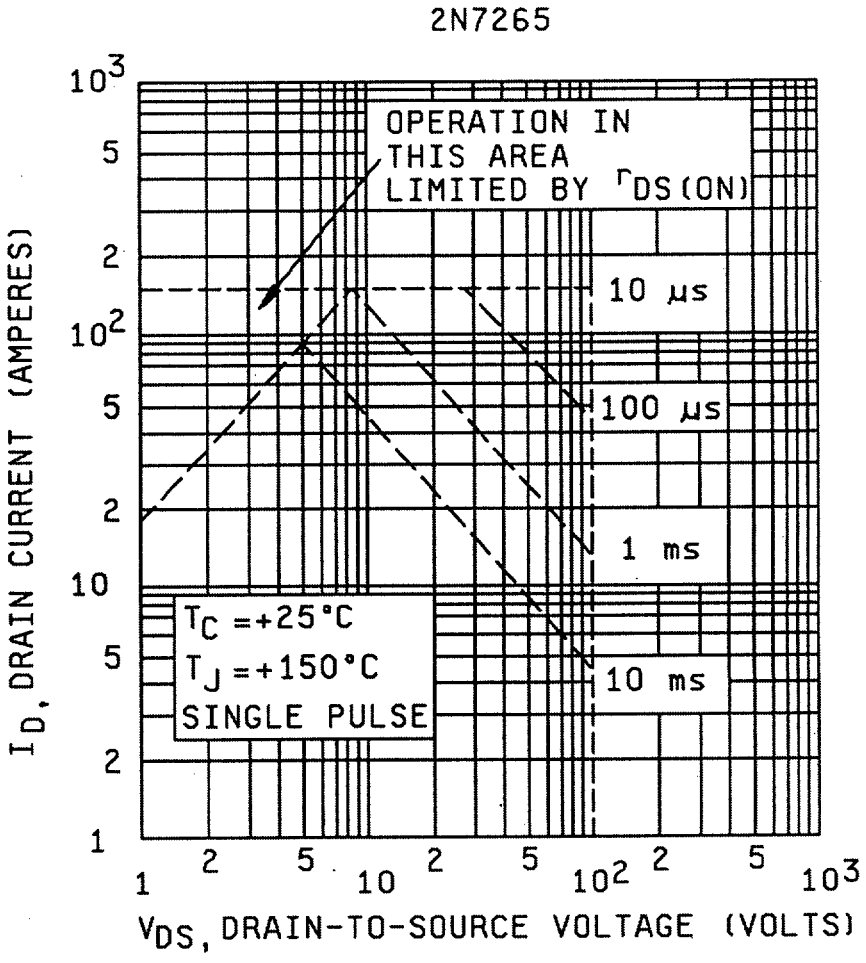


FIGURE 3. Safe operating area graph (2N7265).

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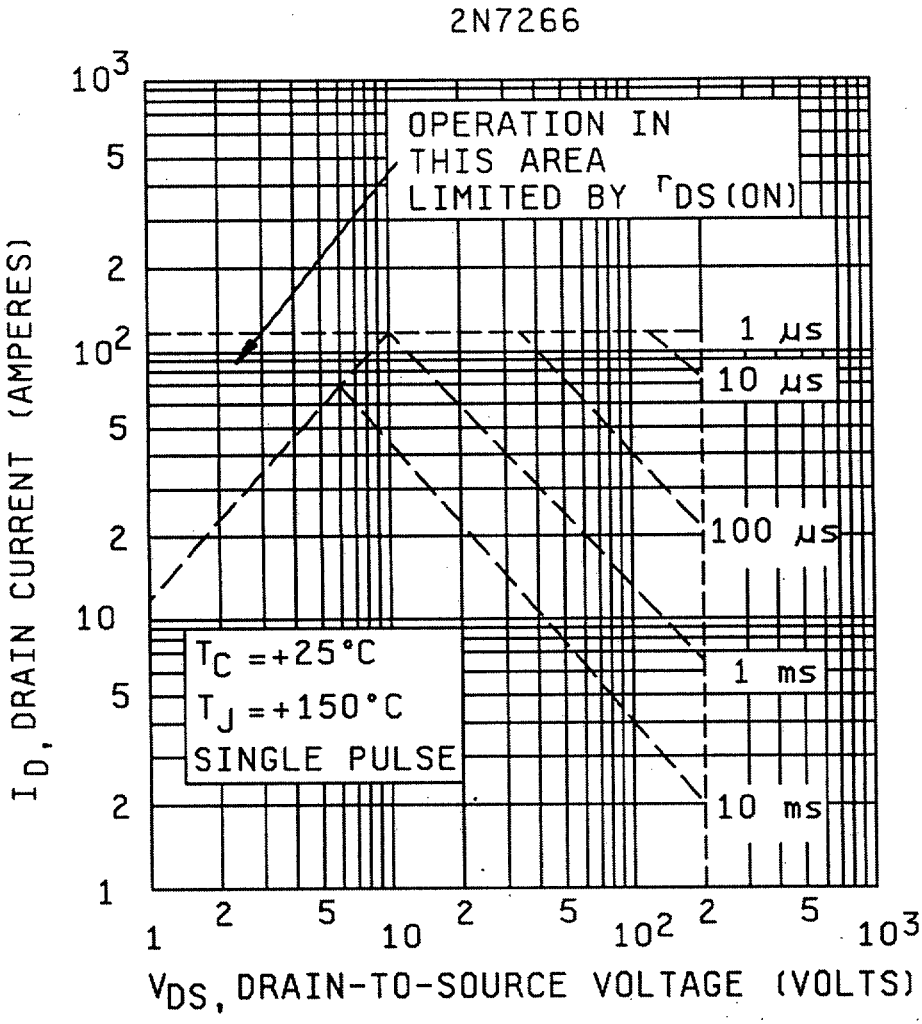


FIGURE 4. Safe operating area graph (2N7266).

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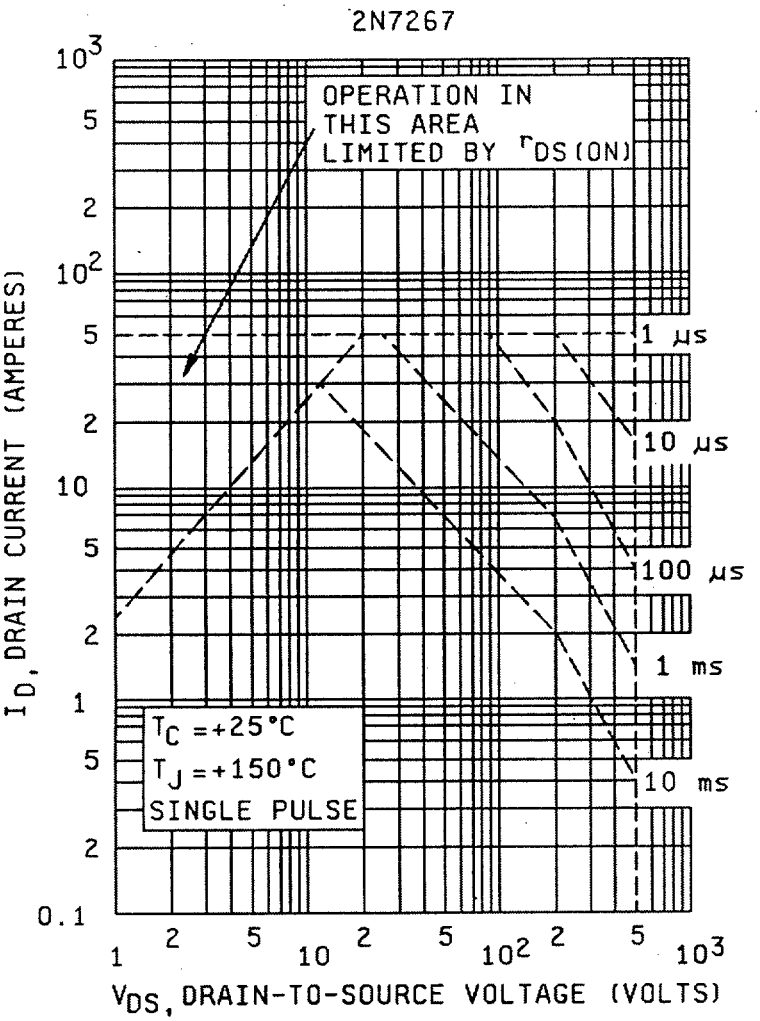


FIGURE 5. Safe operating area graph (2N7267).

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5. PACKAGING

5.1 Packaging. Packaging shall prevent mechanical damage of the devices during shipping and handling and shall not be detrimental to the device. When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points' packaging activity within the Military Department or Defense Agency, or within the Military Departments' System Command. Packaging data retrieval is available from the managing Military Departments' or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

5.2 Marking. Unless otherwise specified (see 6.2), marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. See MIL- PRF-19500.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No.19500 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center Columbus, ATTN: DSCC-VAT, 3990 East Broad Street, Columbus, OH 43216-5000.

6.4 Substitution information. Devices covered by this specification are substitutable for the manufacturer's and user's Part or Identifying Number (PIN). This information in no way implies that manufacturer's PIN's are suitable for the military PIN.

Preferred types	Commercial types
2N7265	IRH7150 1/
	IRH8150
2N7266	IRH7250 1/
	IRH8250
2N7267	IRH7450 1/
	IRH8450

1/ IRH7150, IRH7250, IRH7450 3k, 10k, 100k RAD (Si)
IRH8150, IRH8250, IRH8450 1000k RAD (Si).

6.5 Ordering data. Acquisition documents may specify the material and finish (see 3.3.1).

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CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Navy - TD
Air Force - 19, 70, 80

Preparing activity:
DLA - CC

(Project 5961- 1902-02)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-PRF-19500/602B	2. DOCUMENT DATE (YYMMDD) 970603

3. DOCUMENT TITLE
SEMICONDUCTOR DEVICE, FIELD EFFECT RADIATION HARDENED (TOTAL DOSE ONLY) TRANSISTOR, N-CHANNEL, SILICON TYPES 2N7265, 2N7266, AND 2N7267, JANTXVM, D, R, F, G, AND H, AND JANSM, D, R, F, G, AND H

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Alan Barone	b. TELEPHONE (Include Area Code) (1) Commercial (614)692-0510 (2) AUTOVON 850-0510
c. ADDRESS (Include Zip Code) Defense Supply Center Columbus, ATTN: DSCC-VAT, 3990 East Broad Street, Columbus, OH 43216-5000	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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